

Camera Recording Guide

Recording video from the robot's cameras, watching it back, and changing the settings.

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1 Before you start

Your laptop must be on the same Wi-Fi as the robot (`NECL_5G`). Connect to the robot's onboard computer:

```
ssh eric@sixsevensupremacy.local
```

When prompted, enter the password: `123`

Then open the project:

```
cd ~/tidybot2
conda activate tidybot2
```

Everything below is run from that `~/tidybot2` folder with the `tidybot2` environment active — your prompt will start with `(tidybot2)`.

2 Record a video

```
python record_video.py
```

This opens all three cameras, waits for them to warm up, and starts recording. Press **Ctrl+C** when you want to stop. To record a fixed length instead and stop automatically:

```
python record_video.py --seconds 30
```

Each recording is saved to its own timestamped folder:

```
~/tidybot2/data/demos/20260902T134311/  
  base_image.mp4  
  left_wrist_image.mp4  
  right_wrist_image.mp4  
  data.pkl
```

You get one `.mp4` per camera. The `data.pkl` file holds timing information used later for training — you can ignore it for viewing.

To save this recording somewhere other than the default, add `--endpoint` with a path:

```
python record_video.py --seconds 30 --endpoint /your/folder/path
```

See Section 5 for the full list of settings and one-off flags.

Only one program can use the cameras at a time. If the live monitor (Section 4) is running, stop it first or the recording will fail to open the cameras.

3 Find and watch your recordings

List recordings, newest first:

```
ls -t ~/tidybot2/data/demos
```

Watch in the browser (easiest)

On the mini-PC, play back the most recent recording:

```
python camera_monitor.py --replay
```

Then open `http://sixsevensupremacy.local:8000` in a browser on your laptop. All three cameras play in sync with a timeline scrubber and a play/pause button; the recording's folder is shown at the top. To play a specific recording, give its folder:

```
python camera_monitor.py --replay data/demos/20260902T134311
```

Press **Ctrl+C** on the mini-PC when done.

Copy the video files to your laptop

Run this **on your laptop** (not over SSH) — it grabs the most recent recording and opens the videos:

```
LATEST=$(ssh eric@sixsevensupremacy.local 'ls -t ~/tidybot2/data/demos | head -1')
scp -r eric@sixsevensupremacy.local:"~/tidybot2/data/demos/$LATEST" ~/Downloads/
open ~/Downloads/$LATEST/*.mp4
```

To fetch a specific recording, use its folder name instead of `$LATEST`.

4 Check the cameras before recording

The live monitor shows all three feeds in a browser so you can confirm framing and focus before you record. Start it on the mini-PC:

```
python camera_monitor.py
```

Then, in a browser on your laptop, open:

```
http://sixsevensupremacy.local:8000
```

One feed shows large; the others are thumbnails on the side — click a thumbnail to enlarge it. Each feed is labelled with its location and camera serial number. Press **Ctrl+C** on the mini-PC to stop the monitor when you're done, *before* you record.

The same tool plays back a recording with `--replay` — see Section 3.

If the page won't load, make sure your laptop is on NECL_5G. If the name doesn't work, ask the team for the mini-PC's IP address and use `http://<that-ip>:8000` instead.

5 Change the settings

Every setting lives in one file: `record_video_config.py`. Open it in a text editor on the mini-PC:

```
nano ~/tidybot2/record_video_config.py
```

SETTING	WHAT IT CONTROLS
DATA_DIR	Where recordings are saved. Default <code>"data/demos"</code> (inside the project folder).
CAMERAS	Which cameras to record. Put a <code>#</code> at the start of a line to skip that camera.
BASE_CAMERA_SERIAL LEFT_WRIST_CAMERA_SERIAL RIGHT_WRIST_CAMERA_SERIAL	The serial numbers that identify each camera. Get current ones by running <code>python -m cameras</code> .
RECORD_HZ	How many frames per second are saved. Default <code>10</code> .
LOGITECH_RESOLUTION REALSENSE_RESOLUTION	Video size as <code>(width, height)</code> .
REALSENSE_CAPTURE_FPS	How fast the wrist cameras capture. Must be <code>30</code> , <code>60</code> , or <code>90</code> .
LOGITECH_FOCUS	Base camera focus. <code>0</code> = focus far away; use <code>100</code> if the fisheye lens is fitted.
VIDEO_CODEC	Video format. Leave as <code>"mp4v"</code> unless you know the mini-PC has an H.264 encoder.
MAX_SECONDS	Auto-stop after this many seconds. <code>None</code> means record until you press Ctrl+C.

Save the file (in `nano`: Ctrl+O, Enter, then Ctrl+X). The new settings take effect the next time you run `record_video.py`.

One-off overrides

To change something for a single recording without editing the file:

```
python record_video.py --seconds 30 --endpoint /your/folder/path
```

FLAG	EFFECT
--seconds N	Stop after N seconds.
--endpoint PATH	Save this recording to PATH instead of the default.
--record-hz HZ	Use a different frames-per-second for this recording.

6 If something goes wrong

SYMPTOM	FIX
Recording fails to open a camera / <code>AssertionError</code>	The live monitor is probably still running. Stop it (Ctrl+C on the mini-PC), then try again.
“RealSense ... not found”	Check the wrist camera’s USB cable (it needs a blue USB 3 port). Run <code>python -m cameras</code> to see what’s connected and its serial.
The video is all black	Lens cap still on, or the camera is pointed at nothing. Use the live monitor to check.
The monitor page won’t load	Confirm your laptop is on <code>NECL_5G</code> . Try the mini-PC’s IP address instead of its name.
“command not found: python”	The environment isn’t active. Run <code>conda activate tidybot2</code> first.

Quick reference

<code>ssh eric@sixsevensupremacy.local</code>	connect to the robot computer (password <code>123</code>)
<code>cd ~/tidybot2 && conda activate tidybot2</code>	open the project
<code>python record_video.py</code>	record (Ctrl+C to stop)
<code>python record_video.py --seconds 30</code>	record for 30 seconds
<code>python record_video.py --endpoint PATH</code>	save this recording to PATH
<code>ls -t ~/tidybot2/data/demos</code>	list recordings, newest first
<code>python camera_monitor.py</code>	live monitor → <code>http://sixsevensupremacy.local:8000</code>
<code>python camera_monitor.py --replay</code>	play back the latest recording in the browser
<code>python -m cameras</code>	list connected cameras and serials
<code>nano ~/tidybot2/record_video_config.py</code>	change settings